

Date: 2026-05-07

This scope of work defines the fieldwork, analysis, and deliverables needed to prepare a Soils Specialist Report for the East Crazy Inspiration Divide Land Exchange Environmental Assessment on the Custer Gallatin National Forest. It is written so a consultant, Forest Service soils specialist, or Custer Gallatin interdisciplinary resource specialist can understand what is necessary to complete the soils report and provide report-ready input to the EA.

This SOW is a Project SOW planning and contracting support artifact. It is not a final SOW award document, an applicability decision, a compliance finding, legal advice, a legal sufficiency determination, or a final agency decision.

Source Basis

This SOW was reorganized from current source documents and East Crazies package documents.

Source requirements:

- 36 CFR 220.4 and 36 CFR 220.7 require the EA record to support planning, decision-making, and public disclosure and to include need, proposed action, alternatives, environmental effects, and agencies/persons consulted.
- Region 1 FSM 2550 Soil Management Supplement requires management practices that maintain or improve soil quality. It identifies detrimental soil disturbance (DSD) as compaction, rutting, displacement, severe burning, surface erosion, loss of surface organic matter, and soil mass movement; requires new activities to keep cumulative detrimental soil conditions within the applicable activity-area standard; and recommends photo points, visual methods, transects, and detailed sampling when visual methods are not enough.
- The Custer Gallatin Land Management Plan soil direction includes FW-DC-SOIL-01 for inherent soil productivity, FW-DC-SOIL-02 and FW-DC-SOIL-03 for organic substrates and coarse woody debris, FW-STD-SOIL-01 for detrimental soil conditions, and FW-GDL-SOIL-01 through FW-GDL-SOIL-08 for equipment slope limits, blading depth, restoration methods, scarification depth, landslide avoidance, soil pit/trench handling, coarse woody debris, and burn pile restoration.
- Forest Service Soil Quality Monitoring materials, including the Forest Soil Disturbance Monitoring Protocol and Soil-Disturbance Field Guide, supply the expected soil-disturbance class framework.
- NRCS Web Soil Survey and SSURGO supply the official soil map units, soil properties, interpretations, and GIS data needed for baseline soil conditions.
- National Core BMPs and applicable Montana, Forest Service, and Custer Gallatin BMP direction supply implementation controls for erosion, sediment, drainage, water quality, and invasive species.

Specialist-report model reviewed:

- Soils Report Revised_Anderson 2023.pdf, South Plateau Landscape Area Treatment Project. That report is the local specialist-report model for structure and level of detail. It includes affected environment, applicable Forest Plan standards and guidelines, methodology, spatial and temporal scale, resource indicators, DSD analysis, CWD analysis where vegetation management is involved, direct and indirect effects, cumulative effects, Forest Plan compliance, references, and quantified tables for DSD and mitigation assumptions.

What the local soils report model did:

- identified the affected environment and existing condition before effects analysis;
- stated the applicable Forest Plan standards and guidelines, including DSD and CWD direction;
- defined spatial and temporal scales before comparing alternatives;
- selected resource indicators and indicator measures;
- used field and remote sampling to identify existing DSD;
- calculated proposed and cumulative DSD where an activity area could be defined;
- prescribed mitigation or dropped units where calculated DSD would exceed the applicable threshold;
- analyzed direct, indirect, and cumulative effects by alternative;
- closed with Forest Plan compliance, references, and supporting tables.

This SOW carries that report structure forward, but it does not assume East Crazies requires the same quantified DSD treatment. East Crazies is a land exchange with limited ground disturbance, dominated by trail, trailhead, parcel-restriction, wetland/riparian, access, and sediment-connection questions. The soils specialist must therefore either confirm the existing East Crazies no-meaningful-soil-effect rationale or revise it with field evidence, GIS evidence, and a clear Forest Plan rationale.

East Crazies documents reviewed:

- East Crazy Inspiration Divide Final Environmental Assessment, January 2025.
- East Crazies Plan Consistency Table.
- East Crazies Roads, Trails, and Access Report.
- East Crazies Wetlands Report.
- Post Exchange Federal Parcels and Restrictions.
- East Crazies Project SOW intake fixture and generated Project SOW package smoke output.

East Crazies soils-analysis reality:

- The East Crazies final specialist report folder does not include a standalone soils specialist report.
- The Plan Consistency Table addressed soils directly. It found the only ground-disturbing

activities were the new Sweet Trunk Trail and the Inspiration Divide Trail reroute; stated those activities would not affect soil resources or organic substrates at a large enough scale to be meaningful; treated FW-STD-SOIL-01 as not applicable because trail building was not one of the activities to which that standard was intended to apply and defining the activity area for a linear trail was problematic; and marked most soil guidelines not applicable because the project did not include timber harvest, temporary road construction, blading, vegetation management, reforestation, soil pits/trenches, pile burning, or ground equipment on steep grades.

- The Plan Consistency Table did apply related road/trail and watershed guidelines to trail construction: hydrologically disconnect trail drainage from riparian management zones and water bodies, avoid high mass-wasting potential if found during trail construction, and harden trail or stream crossings to protect streambeds, banks, and approaches.

- The final EA described Sweet Trunk Trail No. 274 as a new 22-mile trail, the Inspiration Divide Trail No. 8 relocation as about 1,500 feet, and Big Timber Canyon Trailhead improvements as potentially including resurfacing, additional parking, toilet facilities, and an interpretive kiosk. The EA also stated trail construction could deliver sediment where the trail nears or crosses waterbodies, but that magnitude and duration would be relatively low with design features and LMP protections; the Trail No. 8 reroute would remove an unnecessary crossing of an unnamed intermittent stream and adjacent wetland.

- The Wetlands Report documented actual fieldwork on September 23-28, 2022. A Custer Gallatin wetland scientist assessed Federal Parcels 1 and 2 in a meandering manner; excavated soil pits; evaluated hydric soil indicators, vegetation, and hydrology; mapped wetland boundaries with resource-grade GPS; used confirmation test pits where full sample points were not collected; and used those field results to calibrate a refined preliminary wetland assessment across the remaining parcels using MTNHP, Google Earth, ESRI imagery, NHD, NWI, professional judgment, and GIS.

- Post-exchange restrictions include no-subdivision/no-mineral-development deed restrictions on Federal Parcels 1-4, a cultural protective covenant on Parcel 6, and wetland protective covenants on Parcels 7-10.

Project Context

- Project: East Crazy Inspiration Divide Land Exchange Proposed Action
- Forest: Custer Gallatin National Forest
- District: Bozeman Ranger District
- NEPA level: environmental assessment
- Project type: land exchange
- Forest plan profile: custer_gallatin
- Resource area: soil_resources
- Selected Project SOW resource scope: vegetation_soils_air_quality
- Current intake status: soil_resources is not explicitly listed as a proposed-action-derived resource area, but soils are implicated by trail construction, trail relocation, trailhead work, wetland/riparian protections, grazing-related infrastructure, parcel restrictions, and sediment-delivery analysis.

The consultant must prepare a standalone Soils Specialist Report unless the Forest Service soils reviewer affirmatively decides that a no-issue/incorporated-analysis memo is sufficient. Either way, the Fieldwork, Analysis, and Deliverables below must be addressed.

Fieldwork

Fieldwork must verify the soils facts needed to support the Soils Specialist Report. It must be scoped to East Crazies conditions and to the soil-resource requirements above.

Fieldwork Objectives

The field effort must answer these questions:

- Where would ground disturbance occur from the Sweet Trunk Trail, Inspiration Divide Trail reroute, Big Timber Canyon Trailhead improvements, easements, wetland/riparian protections, grazing-related improvements, or parcel-management restrictions?
- Are there soil map units, landforms, slopes, hydric soils, unstable areas, compacted areas, eroding areas, or sediment-routing pathways that require design criteria, mitigation, or monitoring?
- Do soil conditions support the Plan Consistency Table conclusion that the project would not affect soil resources or organic substrates at a meaningful scale?
- If FW-STD-SOIL-01 is not applied to the linear trail features, what soil-quality indicators and activity-area rationale explain that decision?
- What BMPs, drainage controls, revegetation requirements, and post-construction monitoring are necessary for trail, trailhead, wetland/riparian, and access-related work?

Pre-Field Data Review

Before field mobilization, compile and review:

- proposed action, alternatives, final EA, Plan Consistency Table, Roads/Trails/Access Report, Wetlands Report, maps, and decision documents;
- Sweet Trunk Trail No. 274 alignment, Inspiration Divide Trail No. 8 reroute, Big Timber Canyon Trailhead improvement footprint, road/trail easements, stream crossings, wetland/riparian areas, and parcel boundaries;
- federal disposal parcels and non-federal acquisition parcels, including post-exchange restrictions and any restrictions that affect soil, wetland, mineral, or development conditions;
- NRCS Web Soil Survey/SSURGO soil map units, hydrologic soil groups, erosion hazard, slope, drainage, compaction/rutting limitations, revegetation limitations, and hydric soil ratings;
- Custer Gallatin soil, watershed, road/trail, riparian, recreation, invasive species, and recommended-wilderness/backcountry plan components that could affect soil work;
- NHD, NWI, MTNHP wetland/riparian data, DEM/slope data, aerial imagery, existing route layers,

grazing allotment and improvement layers, and any available landslide or mass-wasting data;

- Wetlands Report field data and wetland boundary assumptions, especially Federal Parcel 2 delineation results and refined preliminary wetland estimates for the remaining parcels;
- Roads/Trails/Access Report route status, trail classifications, easement tables, and INFRA route information;
- any prior soil, hydrology, road, wetland, range, vegetation, botany, minerals, hazardous-materials, or realty notes that identify soil-related issues.

Fieldwork Plan

Submit a fieldwork plan for Forest Service review before mobilization. The plan must include:

- field questions and decision points;
- activity-area definitions for each soil-relevant action element;
- maps of proposed field review areas, soil map units, route alignments, wetland/riparian areas, stream crossings, slopes, and parcel boundaries;
- proposed observation points, transects, meander routes, representative soil checks, and photo points;
- method for applying Forest Service soil-disturbance classes to trail, trailhead, wetland/riparian, grazing-improvement, and parcel-management conditions;
- method for documenting why FW-STD-SOIL-01 does or does not apply to the linear trail features;
- access needs, landowner coordination needs, safety plan, communications plan, and seasonal constraints;
- field forms, GPS/photo naming conventions, GIS projection, and data-transfer format;
- coordination points with hydrology/wetlands, roads/trails/access, vegetation/botany, range, realty, minerals, hazardous materials, and Forest Plan consistency reviewers.

Field Timing

Fieldwork must occur during a snow-free period when soil surface conditions can be observed. Avoid field conclusions when frozen ground, snow cover, wildfire closure, unsafe access, or saturated conditions prevent reliable evaluation. A targeted wet-season or post-storm revisit must be recommended if runoff, drainage, wetland connectivity, erosion, or sediment delivery cannot be evaluated during the primary field visit.

Field Methods

At minimum, fieldwork must include:

- meander review of the Sweet Trunk Trail alignment, Inspiration Divide Trail reroute, and Big Timber Canyon Trailhead improvement area where access allows;
- targeted review of stream crossings, wetland/riparian approaches, slope breaks, toe slopes,

drainages, seeps/springs, hydric-soil indicators, and sediment-routing pathways;

- verification of representative NRCS soil map units, slope classes, landforms, surface texture, coarse fragments, drainage, hydric or wet soil indicators, and erosion susceptibility;
- visual soil-disturbance classification for existing compaction, rutting, displacement, puddling, surface erosion, loss of surface organic matter, and mass-movement features;
- GPS/photo documentation of representative soil conditions, existing disturbance, sensitive features, stream/wetland crossings, drainage concerns, erosion features, and proposed disturbance footprints;
- shovel checks, soil probes, or soil pits where needed to verify hydric indicators, rooting depth, texture, restrictive layers, compaction, wetness, or restoration feasibility;
- review of wetland confirmation-test-pit assumptions from the Wetlands Report where soil/wetland boundaries affect trail, deed restriction, or sediment-delivery analysis;
- review of whether proposed trail design can meet hydrologic-disconnection, mass-wasting-avoidance, and hardened-crossing plan direction;
- review of soil conditions on federal parcels leaving ownership and non-federal parcels entering ownership where parcel access and project purpose require it.

Detailed lab sampling is not required unless the Forest Service soils reviewer determines a specific question cannot be resolved by visual methods, soil survey, field checks, and professional judgment.

Minimum Field Record

For each activity area or representative field check, record:

- date, observers, weather, soil moisture, and access limitations;
- activity-area ID and action element;
- GPS point, track, transect, or polygon identifier;
- photo IDs and captions;
- soil map unit and field verification notes;
- slope, landform, drainage position, hydric/wet soil indicators, and sediment-connectivity notes;
- existing disturbance type and soil-disturbance class;
- expected disturbance source and approximate footprint, length, or acreage;
- erosion, compaction, rutting, displacement, mass movement, organic matter, and ground-cover observations;
- applicable BMP, design feature, mitigation, or monitoring need;
- uncertainty, data gap, or revisit need.

Analysis

Analysis must translate the field record and source documents into report-ready soil findings. It must use the South Plateau soils report as the local structure model while staying tailored to East Crazies.

Analysis Framework

The Soils Specialist Report must include:

- affected environment and existing condition;
- applicable Forest Plan standards and guidelines;
- methodology, including spatial and temporal scale;
- resource indicators and indicator measures;
- direct and indirect effects;
- cumulative effects;
- mitigation, BMPs, and monitoring;
- Forest Plan consistency;
- references and administrative-record index.

Spatial And Temporal Scale

Define and justify spatial and temporal bounds.

Spatial bounds:

- Direct effects: Sweet Trunk Trail, Inspiration Divide Trail reroute, Big Timber Canyon Trailhead, access/easement work, stream/wetland approaches, restoration or revegetation sites, and any construction staging areas.
- Indirect effects: connected slopes, drainage paths, riparian management zones, wetlands, stream crossings, road/trail drainage networks, and sediment-routing areas.
- Cumulative effects: project boundary plus any watershed, route, allotment, parcel, or wetland context needed to evaluate additive soil disturbance and sediment effects from past, present, and reasonably foreseeable actions.

Temporal bounds:

- Existing condition: current field-observed and GIS/record condition.
- Short term: construction through stabilization/revegetation of disturbed soil.
- Long term: period during which trail use, trailhead use, parcel management, erosion, compaction, drainage, or revegetation could affect soil function.

Resource Indicators

Use these indicators unless the Forest Service soils reviewer approves changes:

- DSD or equivalent soil-disturbance indicator for compaction, rutting, displacement, puddling, surface erosion, loss of surface organic matter, severe burning if relevant, and mass movement.

- Soil productivity and long-term site productivity.
- Hydrologic soil function, infiltration, drainage, and sediment delivery.
- Organic substrates, litter/duff, soil organic matter, and coarse woody debris where vegetation or forested activity areas are involved.
- Wet/hydric soil and riparian soil condition where trail, wetland, or stream-adjacent actions occur.
- Revegetation and invasive-species establishment risk.

The report must explicitly state whether FW-STD-SOIL-01 is applied, modified for linear trail features, or screened out. If screened out, the report must still evaluate soil quality using the indicators above and must explain why the Plan Consistency Table’s earlier rationale remains valid or needs revision.

Existing Condition Analysis

Describe:

- soil map units, landforms, slopes, parent materials, hydrologic soil groups, drainage, erosion hazard, compaction/rutting limitations, revegetation limitations, hydric-soil indicators, and sensitive soil attributes;
- existing roads, trails, trailheads, stream crossings, grazing improvements, disturbed areas, wetland/riparian areas, and route-drainage conditions;
- existing soil disturbance in activity areas and representative parcels;
- soil conditions on federal parcels leaving ownership and non-federal parcels entering ownership, where relevant to the soil report and land exchange decision;
- current soil-water connectivity and sediment-routing pathways;
- data gaps and limits of confidence.

Direct And Indirect Effects Analysis

Analyze no action, proposed action, Alternative 1 if retained for analysis, and any other alternative provided by the Forest Service.

The analysis must address:

- soil disturbance from the new Sweet Trunk Trail, Inspiration Divide Trail reroute, Big Timber Canyon Trailhead improvements, easements, stream crossings, staging, and restoration;
- whether trail construction near or across waterbodies could deliver sediment, and whether magnitude and duration remain low with design features and LMP protections;
- whether the Inspiration Divide Trail reroute reduces soil/sediment effects by eliminating the existing unnecessary stream and wetland crossing;
- whether wetland deed restrictions and retained wetland parcels adequately protect wet/hydric soils, riparian soils, and soil-water functions;

- whether no-subdivision/no-mineral-development restrictions and wetland protective covenants reduce foreseeable soil disturbance on federal parcels leaving ownership;
- whether acquired parcels create new soil-management needs for NFS management;
- whether trailhead resurfacing, additional parking, toilet facilities, or kiosk construction need erosion-control, compaction, drainage, topsoil, or revegetation design criteria;
- whether grazing-related improvements or continued grazing affect compaction, bare soil, bank disturbance, erosion, or revegetation needs;
- whether hazardous-materials or mineral conditions create soil handling, avoidance, disclosure, or mitigation needs;
- whether restoration, invasive-species prevention, carbon, or vegetation assumptions rely on soil conditions.

Quantify expected disturbance by action element, activity area, soil map unit, and alternative when data allow. For linear features where acreage-based DSD is not useful, disclose length, width, crossing count, sensitive-soil crossings, wetland/riparian approaches, drainage features, and representative soil-disturbance class instead.

Cumulative Effects Analysis

Evaluate cumulative soil effects only where there is a useful cause-and-effect relationship. Screen:

- existing and proposed routes, trails, trailheads, unauthorized routes, and route drainage;
- grazing use and range improvements;
- wetland/riparian protections or disturbance;
- foreseeable trail construction, road maintenance, restoration, or recreation improvements;
- private land development risk on parcels leaving federal ownership, considering deed restrictions and protective covenants;
- vegetation, invasive species, fire/fuels, carbon, and restoration actions that could alter ground cover, erosion, or revegetation.

Mitigation, BMPs, And Monitoring Analysis

Prepare a mitigation/BMP/monitoring matrix. Each row must include action element, soil concern, location trigger, required measure, responsible party, timing, and verification method.

At minimum, analyze:

- hydrologic disconnection of trail drainage from riparian management zones and waterbodies;
- drainage spacing, rolling dips, grade reversals, waterbars, cross drains, armoring, outsloping, and sediment filtering;
- stream-crossing hardening for trail approaches, streambeds, banks, and approaches;
- avoidance or design modification for high mass-wasting potential, wet soils, hydric soils, steep

- slopes, unstable areas, and sensitive soil map units;
- construction timing and wet-weather restrictions;
 - erosion and sediment control during trail, trailhead, and access work;
 - topsoil/duff salvage, decompaction, revegetation, mulch, weed-free seed, weed prevention, and restoration;
 - photo-point or post-construction monitoring for trail, trailhead, wetland/riparian, and revegetation areas;
 - whether any DSD or soil-disturbance monitoring is needed even if FW-STD-SOIL-01 is not applied to trail building.

Forest Plan Consistency Analysis

Evaluate and document consistency with:

- FW-DC-SOIL-01, FW-DC-SOIL-02, FW-DC-SOIL-03;
- FW-STD-SOIL-01, including the rationale for applicability or non-applicability;
- FW-GDL-SOIL-01 through FW-GDL-SOIL-08;
- road/trail guidelines applied in the Plan Consistency Table, especially FW-GDL-RT-03, FW-GDL-RT-06, and FW-GDL-RT-07;
- invasive-species and revegetation direction, including Forest Service standard operating procedures and BMPs for equipment/material cleaning and prompt revegetation after ground-disturbing management activities;
- riparian, wetland, aquatic, recreation, recommended-wilderness, backcountry, and designated-area components where they affect soil or sediment analysis.

The final report must explain where it agrees with, narrows, or revises the existing Plan Consistency Table.

Deliverables

Deliverables must be organized so the Forest Service can place them directly in the administrative record and use them in EA text, Forest Plan consistency review, and implementation.

Deliverable 1: Kickoff And Data Request Log

Provide:

- kickoff meeting notes;
- information received and missing;
- source documents reviewed;
- access constraints;
- stop conditions;

- Forest Service decisions needed before fieldwork.

Acceptance criteria:

- identifies missing GIS, parcel, trail design, wetland, route, or Forest Plan data;
- identifies whether the soil report will proceed as standalone report or no-issue memo pending field verification;
- identifies field-season and access constraints.

Deliverable 2: Fieldwork Plan

Provide:

- field objectives;
- activity-area definitions;
- field maps;
- proposed routes, transects, observation points, soil checks, wetland/riparian checks, and photo points;
- field forms;
- safety/access plan;
- data naming conventions;
- schedule;
- coordination needs.

Acceptance criteria:

- covers Sweet Trunk Trail, Inspiration Divide reroute, Big Timber Canyon Trailhead, stream/wetland interfaces, and parcel restrictions;
- identifies how soil disturbance will be classified for linear trail features;
- identifies whether additional wet-season or post-storm verification may be needed.

Deliverable 3: Field Data Package

Provide:

- GPS points, tracks, polygons, and activity-area refinements;
- completed field forms;
- geotagged photo log;
- soil check notes and soil map unit verification;
- wetland/riparian and sediment-connectivity notes;
- field findings memo identifying urgent design, BMP, mitigation, or monitoring issues.

Acceptance criteria:

- every material field conclusion is traceable to location, date, observer, photo, and activity area;
- sensitive soil, hydric/wet soil, erosion, compaction, rutting, displacement, drainage, and sediment-delivery concerns are mapped or explicitly screened out;
- data are suitable for Custer Gallatin GIS/ID-team use.

Deliverable 4: Analysis Tables And Map Atlas

Provide:

- soil map unit and activity-area crosswalk;
- existing disturbance and expected disturbance table;
- action-element by soil concern matrix;
- stream/wetland crossing and sediment-connectivity table;
- mitigation/BMP/monitoring matrix;
- Forest Plan consistency matrix;
- data-gap and uncertainty table;
- project location, soil-analysis area, soil map unit, slope/erosion hazard, wetland/riparian, road/trail/trailhead, field observation, and mitigation maps.

Acceptance criteria:

- maps include scale, north arrow, coordinate system, data date, project boundary, activity-area IDs, and source attribution;
- tables distinguish no action, proposed action, and Alternative 1 or other alternatives;
- linear-feature effects are disclosed with useful measures, not forced into an acreage metric where that would mislead.

Deliverable 5: Draft Soils Specialist Report

Provide a complete draft report with:

- executive summary;
- project description and alternatives;
- fieldwork and methods;
- affected environment;
- direct and indirect effects;
- cumulative effects;
- mitigation, BMPs, and monitoring;
- Forest Plan consistency;
- unavoidable adverse effects and productivity considerations if relevant;

- references and administrative-record index;
- appendices for field forms, photos, GIS layers, DSD/soil-disturbance calculations, and maps.

Acceptance criteria:

- follows the level of detail expected from the South Plateau soils specialist report while remaining tailored to East Crazies;
- explains whether a standalone soils report is required or whether a no-issue/incorporated-analysis memo is sufficient;
- explicitly addresses what East Crazies already did in the Plan Consistency Table, Roads/Trails Report, Wetlands Report, and final EA;
- ties every soil conclusion to field evidence, GIS evidence, source documents, or specialist judgment.

Deliverable 6: Draft EA Soils Section

Provide concise EA-ready text for:

- affected environment;
- environmental consequences;
- mitigation/design criteria;
- Forest Plan consistency;
- cumulative effects if needed.

Acceptance criteria:

- suitable for insertion into an EA or appendix;
- references the soils specialist report rather than duplicating all technical detail;
- uses plain language for public disclosure.

Deliverable 7: Final Soils Specialist Report

Provide:

- final report;
- final PDF;
- editable source file;
- comment-response or change log;
- final tables, maps, appendices, and administrative-record index.

Acceptance criteria:

- resolves Forest Service and ID-team comments;
- preserves source citations and field-data traceability;
- clearly states residual uncertainty and Forest Service decisions;
- final PDF starts with a valid %PDF- header.

Deliverable 8: Final Data Package

Provide:

- GIS layers or geopackage/file geodatabase;
- PDF map atlas;
- photo log and photos;
- field forms;
- source data inventory;
- metadata/readme;
- data limitations memo.

Acceptance criteria:

- GIS and table outputs use agreed naming, projection, and metadata;
- the Forest Service can reproduce report maps and trace field conclusions;
- source-library or administrative-record file paths are documented.

Requirements Traceability

Source document or evidence Fieldwork requirement Analysis requirement Deliverable requirement
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Region 1 FSM 2550 Soil Management Supplement Observe and document compaction, rutting, displacement, erosion, organic-matter loss, severe burning if relevant, and mass movement; use photo points, visual methods, transects, or detailed sampling as needed.
Decide whether DSD standards apply; if they do, evaluate cumulative detrimental soil conditions against the applicable activity area; if they do not, explain the soil-quality indicator substitute. Include field forms, photo log, activity-area rationale, soil-disturbance findings, and DSD or substitute-indicator rationale in the report and appendices.
South Plateau Soils Specialist Report model Use field and remote review to identify existing disturbance and conditions before calculating or screening effects. Follow the local specialist-report structure: affected environment, methods, scale, indicators, direct/indirect effects, cumulative effects, Forest Plan compliance, and references. Produce a draft and final Soils Specialist Report with report-ready tables, maps, DSD or substitute-indicator calculations, mitigation assumptions, and references.
East Crazies Final EA Field-check the Sweet Trunk Trail, Inspiration Divide Trail reroute, Big Timber Canyon Trailhead, crossings, drainage approaches, and potential sediment-delivery locations. Test the EA's low-magnitude, short-duration sediment and soil-effect conclusions against field/GIS evidence and alternatives. Provide EA-ready soils text and a finding that either supports or revises the current EA treatment.
East Crazies Plan Consistency Table Verify the factual basis for treating trail construction and trail reroute as the only

meaningful ground-disturbing actions. | Evaluate FW-DC-SOIL-01 through FW-DC-SOIL-03, FW-STD-SOIL-01, FW-GDL-SOIL-01 through FW-GDL-SOIL-08, FW-GDL-RT-03, FW-GDL-RT-06, FW-GDL-RT-07, and invasive/revegetation direction. | Provide a Forest Plan consistency matrix and a written explanation where the soils report agrees with, narrows, or revises the existing table. |

| East Crazies Wetlands Report | Coordinate with wetland field data; verify hydric/wet soil indicators, wetland/riparian approaches, confirmation-test-pit assumptions, and sediment connectivity where soil conclusions depend on them. | Analyze hydric soils, wetland protective covenants, soil-water function, trail/wetland crossing risk, and the Trail No. 8 reroute's wetland/stream benefit. | Include wetland/riparian soil maps, soil-check notes, field photos, and cross-references to wetland delineation assumptions. |

| East Crazies Roads, Trails, and Access Report | Verify route alignments, trail classifications, easements, trailhead disturbance, stream crossings, and INFRA/access assumptions that affect soils. | Analyze trail and trailhead disturbance using useful linear-feature measures such as length, width, crossing count, drainage features, and sensitive-soil crossings. | Provide route/soil crosswalks, crossing and drainage tables, map atlas layers, and implementation BMP recommendations. |

| Post Exchange Federal Parcels and Restrictions | Confirm whether deed restrictions, wetland covenants, cultural covenants, access reservations, and parcel management assumptions affect field-review scope. | Evaluate whether restrictions reduce foreseeable private-land soil disturbance and whether acquired parcels create new NFS soil-management needs. | Include a parcel-restriction soil-effects table and administrative-record citations. |

| East Crazies final specialist report folder | Record that no standalone East Crazies soils specialist report was found in the observed final specialist report folder. | Treat existing soils work as incorporated plan-consistency/EA analysis unless the Forest Service requires a standalone report. | Include either a standalone Soils Specialist Report or a no-issue/incorporated-analysis memo with the same fieldwork and analysis traceability. |

Schedule And Review Points

Default schedule:

- Week 1: kickoff, source review, information-request log.
- Week 2: fieldwork plan and Forest Service review.
- Weeks 3-5: fieldwork during suitable soil-observation conditions.
- Week 6: field findings memo and refined analysis plan.
- Weeks 7-9: analysis tables, maps, and draft Soils Specialist Report.
- Weeks 10-11: Forest Service and ID-team review.
- Week 12: final report, final PDF, and final data package.

Review gates:

- Forest Service approves fieldwork plan before mobilization.
- Forest Service reviews field findings before draft report production.
- Forest Service reviews draft report before draft EA text is finalized.
- Forest Service accepts final report before the decision record relies on soil mitigation, design criteria, or monitoring commitments.

Open Forest Service Decisions

The Forest Service must resolve:

- whether soil_resources should be added as an explicit East Crazies proposed-action resource area;
- whether a standalone soils specialist report is required or whether an incorporated/no-issue memo is sufficient;
- whether FW-STD-SOIL-01 is not applicable to the trail work, and what alternative soil-quality indicator language should be used for the report;
- whether any field review is needed on both federal disposal and non-federal acquisition parcels;
- which action alternatives must be analyzed beyond no action, proposed action, and Alternative 1;
- which BMPs, mitigation, design criteria, and monitoring commitments must be carried into the EA, decision record, collection agreement, trail design package, or implementation file;
- which GIS/data package format is required by the Custer Gallatin ID team.

Source Links

- 36 CFR 220.4, Forest Service NEPA general requirements:
<https://ecfr.io/Title-36/Section-220.4>
- 36 CFR 220.7, Environmental assessment and decision notice:
<https://ecfr.io/Title-36/Section-220.7>
- Custer Gallatin Land Management Plan page:
<https://www.fs.usda.gov/r01/custergallatin/planning/forest-plan/custer-gallatin-land-management-plan-forest-plan-revision>
- Custer Gallatin Land Management Plan PDF:
<https://www.fs.usda.gov/media/51590>
- Forest Service Soil Quality Monitoring resources:
<https://www.fs.usda.gov/soils/monitoring.shtml>
- Forest Service Soil-Disturbance Field Guide:
https://www.fs.usda.gov/t-d/php/library_card.php?p_num=0819+1815P
- NRCS Web Soil Survey:
<https://www.nrcs.usda.gov/resources/data-and-reports/web-soil-survey>
- Forest Service National Best Management Practices Program:
<https://www.fs.usda.gov/naturalresources/watershed/bmp.shtml>